Optimal growth model, finite horizon

Main objective: to develop the tools for characterizing dynamic decisions in a model with many periods.

Examples: financial savings, investment in human capital, investment in technology etc.

$\Rightarrow$ abstract from the static choices (e.g. consumption vs. leisure), focus on the dynamic tradeoff only

The environment:

Consider a (representative) consumer who lives for $T+1$ periods, has access to production technology that utilizes capital, and needs to allocate the resources between consumption & capital investm. in each period.

You can think of this as a Robinson-Crusoe economy. OR as a decision problem of the Social Planner

Preferences: $U(c_0, c_1, \dots, c_T)$ - additively separable

$$= \sum_{t=0}^{T} \beta^t u(c_t)$$, where $\beta \in (0,1)$: time discount factor (patience)

Typical assumptions on $u(c)$:

- $u(c)$ - cont. differentiable
- $u'(c) > 0$ (increasing)
- $u''(c) < 0$ (concave)
- $\lim_{c \to 0} u'(c) = +\infty$ ($\Rightarrow$ people never choose $c=0$)

Technology:

- production $F(k_t, N)$, assume $N=1$ (normalize)

  consumer = producer, decides how much k_t to use

  - assume $k_{t+1} \geq 0$, $t = 0, \dots, T$

Typical assumptions on $F(k,N)$:

- $F(k,N)$ - cont. differ.
- $F(0,N) = 0$
- $F_K(k,N) > 0$ (increasing)
- $F_{kk}(k,N) < 0$ (concave)

Sometimes we will assume that $F(k,1) = Ak$, e.g. the las assumption is violated

- $c_t + i_t \leq F(k_t, N)$ (feasibility constraint for the consumer)
- $k_{t+1} \leq i_t + (1-\delta) k_t$ (low of motion for capital)

$$\Rightarrow c_t + \cancel{i_t} + k_{t+1} \leq \cancel{i_t} + (1-\delta) k_t + F(k_t, N)$$

$$\Rightarrow \boxed{c_t + k_{t+1} \leq (1-\delta) k_t + F(k_t, N), \quad t = 0, \dots, T}$$

feasibility constraint / resourse constraint / budget constraint

- k_0 is given

We will label $F(k,N) + (1-\delta) k = f(k)$

Consumer's objective:

- choose the consumption and capital streams $\{c_t, k_{t+1}\}_{t=0}^{T}$
- in order to maximize life-time utility
- subject to the feasibility constraint(s)
  (the tradeoffs imposed by the available technology)

Solving the OGM in finite horizon

Thus, the consumer's decision problem is:

$$\boxed{\begin{aligned} &\max_{\{c_t, k_{t+1}\}_{t=0}^{T}} \sum_{t=0}^{T} \beta^t u(c_t) \\ &\text{s.t.} \quad c_t + k_{t+1} \le f(k_t) \\ &\qquad\; k_{t+1} \ge 0 \\ &\qquad\; k_0 \text{ is given} \end{aligned}}$$

$(\lambda_t \beta^t)$

$(\mu_t \beta^t)$

$= S \subset \mathbb{R}^{T+1}$ $= \{(c_0, .., c_T):$ these constr. are satisfied$\}$

① • The objective f-n is continuous

• the choice set (specified by all the constraints) is compact

$\Rightarrow$ / Weierstrass's Theorem / $\Rightarrow$ the max. of the objective f-n is achieved in this set

$\left(\exists \; \{c_t^*\}_{t=0}^{T} \subset S : \; U(c_0^*, .., c_T^*) = \max\limits_{\{c_t\}_{t=0}^{\infty} \subset S} U(c_0, .., c_T)\right)$

(S: all constr.; all constraints)

One can verify that $f(k)$-concave $\Rightarrow$ S-convex

② $U(c_0, .., c_T)$ - strictly concave $\Rightarrow$ the maximum is unique,

the solution can be obtained by the Kuhn-Tucker method

③ $\lim\limits_{c \to 0} u'(c) = +\infty \Rightarrow c_t^* > 0, \; t = 0, .., T$

(i.e. the solution is interior w.r.t. c_t)

Let's find the solution!

$$\mathcal{L} = \sum_{t=0}^{T} \beta^t \left[u(c_t) - \lambda_t \left(c_t + k_{t+1} - f(k_t)\right) + \mu_t k_{t+1} \right]$$

FOCs: (c_t): $\quad u'(c_t) - \lambda_t = 0 \quad , t = 0 .. T \qquad (1)$

(k_{t+1}): $\quad -\lambda_t + \mu_t + \beta \lambda_{t+1} f'(k_{t+1}) = 0, \; t = 0, .., T-1 \qquad (2)$

(k_{T+1}): $\quad -\lambda_T + \mu_T = 0 \qquad (3)$

K-T cond: $\quad \lambda_t (c_t + k_{t+1} - f(k_t)) = 0 \quad , t = 0, .., T \qquad (4)$

$\mu_t \, k_{t+1} = 0 \quad , \; t = 0, .., T \qquad (5)$

$\lambda_t \geq 0$

$\mu_t \geq 0 \qquad , t = 0, .., T$

$c_t + k_{t+1} \leq f(k_t)$

$k_{t+1} \geq 0$

Key properties of the solution:

① $k_{T+1} = 0$ (no idle capital is left in the last period)

$\ulcorner \; u'(c_t) > 0 \Rightarrow /(1), t = T/ \Rightarrow \lambda_T > 0 \Rightarrow$

$/(3)/ \Rightarrow \mu_T > 0 \Rightarrow /(5), t = T/ \Rightarrow k_{T+1} = 0 \; \lrcorner$

② $c_t + k_{t+1} = f(k_t)$ (the resourse constraint holds with equality – no resources are wasted – not surprising)

$\ulcorner \; u'(c_t) > 0 \Rightarrow /(1)/ \Rightarrow$

$\Rightarrow \lambda_t > 0 \Rightarrow /(4)/ \Rightarrow c_t + k_{t+1} = f(k_t) \; \lrcorner$

③ $\mu_t = 0 \; , \; t = 0, \dots, T-1$

in order to have $c_t > 0$, it must be that $\underset{t=0,..,T}{k_t} > 0 \Rightarrow$

$\Rightarrow \mu_t = 0 \; , \quad t = 0, .., T-1$ $\qquad$ $f(0) = 0$ is key here

Thus, $(1) \Rightarrow u'(c_t) = \lambda_t \; , \quad t = 0, \dots, T$

$$u'(c_{t+1}) = \lambda_{t+1}, \quad t=0,\dots,T-1$$

$$(2) \Rightarrow \quad \lambda_t = \beta \lambda_{t+1} \cdot f'(k_{t+1}) + \mu_t, \quad t=0,\dots,T-1$$

$$③ : \mu_t = 0$$

Euler Equation

$$\Rightarrow (*) \quad \boxed{\underbrace{u'(c_t)}_{\text{marginal utility of current cons.}} = \underbrace{\beta\, u'(c_{t+1})}_{\text{PV of the marginal utility of future cons.}} \cdot f'(k_{t+1})}, \quad t=0,\dots,T-1$$

$f'(k_{t+1})$: how many extra units of future cons. can be generated by giving up 1 unit of current cons.

Property ② $\Rightarrow$ $c_t = f(k_t) - k_{t+1}$ $\Rightarrow$

$$(**) \quad \boxed{\begin{array}{l} u'(f(k_t) - k_{t+1}) = \beta\, u'(f(k_{t+1}) - k_{t+2}) \cdot f'(k_{t+1}) \\ k_0 \text{ is given} \\ k_{T+1} = 0 \quad (\text{by property } ①) \end{array}}$$

Thus $\{k_t\}_{t=0}^{T}$ solves the original decision problem if and only if it solves $(**)$.

(Note that $\{k_t\}_0^T$ pins down $\{c_t\}_0^T$:
$c_t = f(k_t) - k_{t+1}$
Thus we can solve for $\{c_t\}_0^T$ or $\{k_t\}_0^T$ only).

$(**)$ is the second-order difference equation.

It has a 2-parameter family of solutions

(e.g. if k_0, k_1 are known $\Rightarrow$ $(**)$ pins down k_2

k_1, k_2 are known $\Rightarrow$ —— " —— k_3 etc.)

This solution is unique (because it is equivalent to the solution to the original problem, which has a unique solution).

But finding this solution is not trivial...
(Note that we know k_0 & $k_{T+1} = 0$, need to find $\{k_1, k_2, .., k_T\}$)